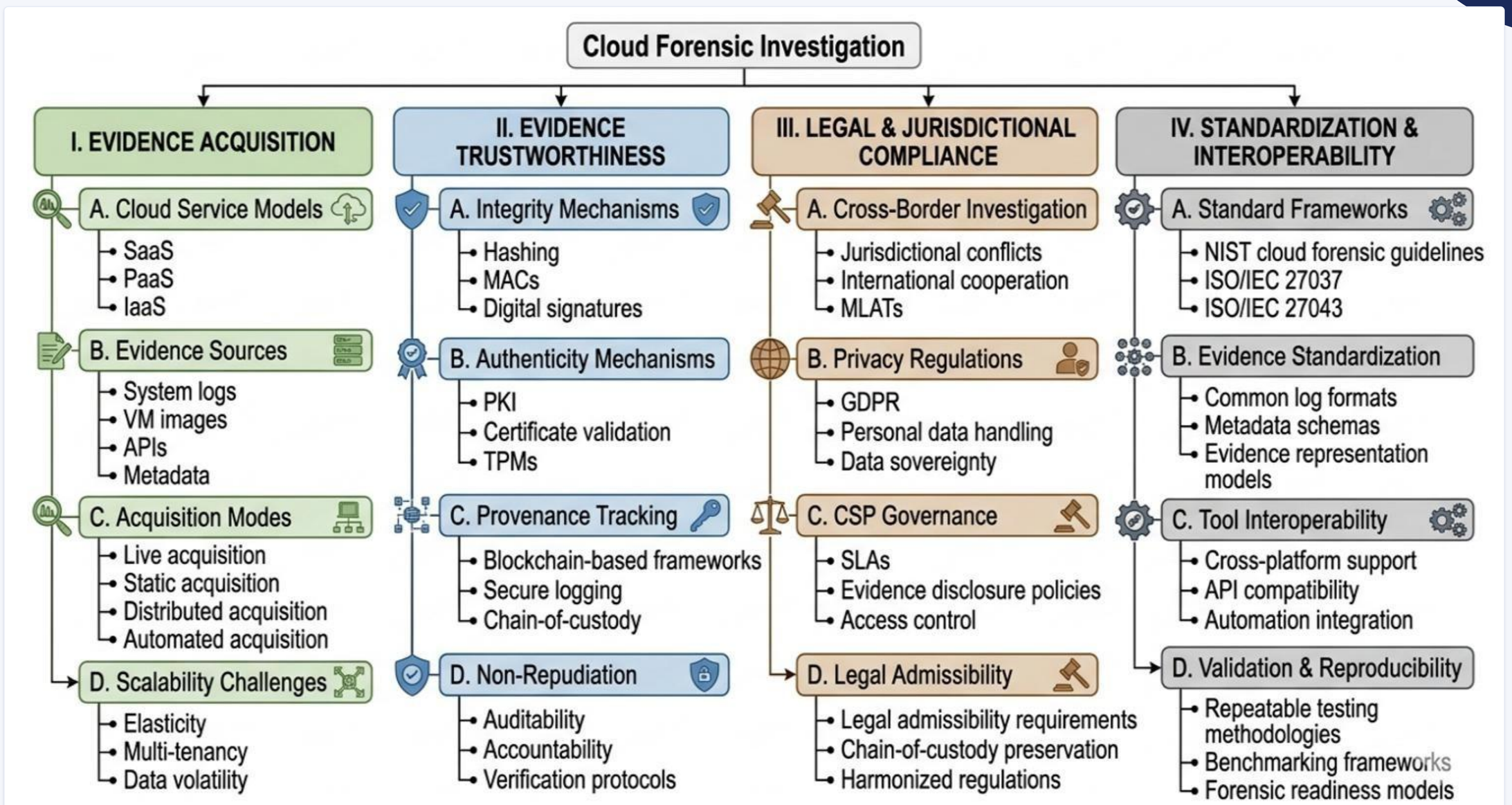
RMTW · FINAL PRESENTATION POSTER

Modern Cloud Forensics

A PRISMA-Guided Systematic Survey of Evidence Acquisition, Integrity, Jurisdictional Challenges, and Standardization Issues

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Four-Dimensional Taxonomy of Cloud Forensic Approaches



PRISMA-guided review: Identification → Screening → Eligibility → Inclusion, applied across four dimensions above.

Findings by Research Question

RQ1
Finding: IaaS dominates forensic acquisition research
Challenge: Limited SaaS/PaaS visibility
Solution: Security Lake, automated acquisition

RQ2
Finding: Blockchain improves evidence integrity
Challenge: Provenance scalability and trust
Solution: Consortium ledgers, immutable logging

RQ3
Finding: GDPR and MLAT processes delay investigations
Challenge: Cross-border admissibility
Solution: Legal readiness frameworks

RQ4
Finding: Poor interoperability across CSPs
Challenge: Log inconsistency and schema mismatch
Solution: OCSF, standardized APIs

Research Gaps

- Limited real-world validation — most studies stop at simulation
- SaaS/PaaS forensics remain underexplored versus IaaS
- No international consensus on forensic standards
- AI's "black box" nature limits legal admissibility
- Conflicting cyber laws slow international cooperation
- Containers, serverless, and edge computing are underexplored

Key Contributions

1. PRISMA-guided, transparent and reproducible survey
2. Four-dimensional taxonomy of cloud forensic approaches
3. Comparative analysis across methodology & contribution
4. Clear map of current limitations across the field

Future Directions

- Explainable AI for court-admissible forensics
- Lightweight blockchain provenance ledgers
- Vendor-neutral, standardized multi-cloud frameworks
- Forensic readiness built into cloud services by design
- Real-time evidencing for containers & serverless
- Privacy-preserving evidence collection

Conclusion

Cloud forensics is shifting from traditional to intelligent approaches — but real progress depends on CSP cooperation, explainable AI, blockchain-based evidence handling, standardized formats, and privacy-preserving collection working together.

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Full list of all 48 references in the complete survey paper. · *mkabir2520016@mscse.uiu.ac.bd*